THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 13: **Statistics** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. For the data with classes 0–20, 20–40, 40–60, 60–80, 80–100, 100–120, 120–140 with frequencies 10, 7, 8, 23, 25, 19, 8 respectively, The mean of the data is: 1

(a) 1100 **(b)** 78 **(c)** 70 **(d)** 77

2. For the data with classes 20–40, 40–60, 60–80, 80–100, 100–120, 120–140, 140–160 with frequencies 21, 11, 5, 16, 25, 9, 13 respectively, The median of the data is: 1

(a) 88.4 **(b)** 96.25 **(c)** 100 **(d)** 80

3. For the data with classes 200–220, 220–240, 240–260, 260–280, 280–300, 300–320, 320–340 with frequencies 3, 21, 7, 17, 16, 16, 20 respectively, The mode of the data is: 1

(a) 240 **(b)** 280 **(c)** 220 **(d)** 231.25

4. The mean of a data is 24 and the assumed mean taken is 20. Then $\frac{\sum f_i d_i}{\sum f_i}$ equals: 1

(a) -4 **(b)** 20 **(c)** 44 **(d)** 4

5. The mean of a distribution with classes 40–60, 60–80, 80–100, 100–120, 120–140, 140–160, 160–180 and frequencies 22, p, 12, 4, 16, 7, 14 respectively is 98.8. The value of p is: 1

(a) 27 **(b)** 24 **(c)** 30 **(d)** 25

6. The median of a distribution with classes 200–220, 220–240, 240–260, 260–280, 280–300, 300–320, 320–340, 340–360 and frequencies 15, 6, 7, x, 16, y, 19, 3 respectively, and total frequency 100, is 290. The value of x is: 1

(a) 22 **(b)** 15 **(c)** 13 **(d)** 14

7. For the data with classes 40–50, 50–60, 60–70, 70–80, 80–90, 90–100, 100–110, 110–120 with frequencies 11, 19, 8, 20, 23, 2, 13, 4 respectively, The mean of the data is: 1

(a) 76.3 **(b)** 85 **(c)** 75.3 **(d)** 941.25

8. For the data with classes 0–20, 20–40, 40–60, 60–80, 80–100, 100–120, 120–140, 140–160 with frequencies 6, 23, 14, 16, 7, 5, 24, 5 respectively, The median of the data is: 1

(a) 60 **(b)** 68.75 **(c)** 76.2 **(d)** 80

9. **Assertion (A):** The mean of grouped data found by the step-deviation method is the same as the mean found by the direct method. 1

Reason (R): The step-deviation method uses $\bar{x} = a + h\,\frac{\sum f_i u_i}{\sum f_i}$ with $u_i = \frac{x_i - a}{h}$, which is a simplified form of the same formula.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** For a grouped distribution with $\sum f_i = 50$, the median class is the class which contains the 25th observation. 1

Reason (R): The mean of grouped data is always equal to its median.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. The ages (in years) of patients admitted to a hospital in a month are given below. **2**

Age (years)	40–50	50–60	60–70	70–80	80–90	90–100	100–110
Number of patients	19	11	8	25	8	14	15

Find the median.

12. The mean of a distribution with classes 40–60, 60–80, 80–100, 100–120, 120–140, 140–160, 160–180, 180–200 and frequencies 16, 25, 16, p, 3, 9, 23, 3 respectively is 107.6. Find the value of p. **2**

13. The weights (in kg) of students of a class are given below. **2**

Weight (kg)	100–120	120–140	140–160	160–180	180–200	200–220	220–240	240–260
Number of students	2	18	7	9	20	15	10	19

Find the mode.

14. The weights (in kg) of students of a class are given below. **2**

Weight (kg)	0–20	20–40	40–60	60–80	80–100	100–120	120–140
Number of students	10	7	8	23	25	19	8

Find the median.

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The daily electricity consumption (in units) of households in a colony is given below. **3**

Units consumed	20–40	40–60	60–80	80–100	100–120	120–140	140–160
Number of households	21	11	5	16	25	9	13

Find the mean and the mode (in this order).

16. The daily electricity consumption (in units) of households in a colony is given below. **3**

Units consumed	0–20	20–40	40–60	60–80	80–100	100–120	120–140
Number of households	22	25	8	17	7	17	4

Find the mean.

17. The median of a distribution with classes 40–60, 60–80, 80–100, 100–120, 120–140, 140–160, 160–180, 180–200 and frequencies 3, 4, 16, x, 16, y, 18, 5 respectively, and total frequency 100, is 127.5. Find the value of x. **3**

OR

The lifetimes (in hours) of electric bulbs tested in a laboratory are given below. **3**

Lifetime (hours)	0–20	20–40	40–60	60–80	80–100	100–120	120–140	140–160
Number of bulbs	6	23	14	16	7	5	24	5

Find the mean and the mode (in this order).

18. The lifetimes (in hours) of electric bulbs tested in a laboratory are given below. **3**

Lifetime (hours)	200–220	220–240	240–260	260–280	280–300	300–320	320–340
Number of bulbs	3	24	17	10	15	14	17

Find the mean.

OR

The median of a distribution with classes 200–220, 220–240, 240–260, 260–280, 280–300, 300–320, 320–340 and frequencies 13, 24, x, 12, y, 8, 15 respectively, and total frequency 100, is 265. Find the value of x. **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** The marks (out of 100) of 40 students in a test are: 0-20: 5 students, 20-40: 8, 40-60: 12, 60-80: 10, 80-100: 5. Take assumed mean $a = 50$ and class size $h = 20$ and let $u_i = \dfrac{x_i - a}{h}$. **5**

(i) Find the class mark of the class 20-40.

(ii) Find u_i for the class 60-80.

(iii) Find $\sum f_i u_i$.

(iv) Find the mean marks by the step-deviation method.

20. The mean of a distribution with classes 20–30, 30–40, 40–50, 50–60, 60–70, 70–80, 80–90, 90–100 and frequencies 2, 22, 5, 20, p, 16, 23, 10 respectively is 63.8. Find the value of p. **5**

OR

The ages (in years) of patients admitted to a hospital in a month are given below. **5**

Age (years)	40–60	60–80	80–100	100–120	120–140	140–160	160–180	180–200
Number of patients	22	12	13	25	5	2	8	13

Find the mean.

END OF QUESTION PAPER — Total: 40 Marks